Ayush Bansal

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Education

The LNM Institute of Information Technology **Aug. 2024 – Jun 2028**
Bachelor of Technology, Electronics and Communication Engineering *Jaipur, India*

Achievements

ETHGlobal Mumbai 2026 – 2nd Place, 0G Track ($1,500) **Nov 2026**
Built SAKA, an autonomous AI DeFi agent on 0G Private Computer, Uniswap & Privy Showcase | GitHub | Demo | X Post

LNMHacks 8.0 – Partner Prize Winner ($600) **Jan 2026**
Won Viral Consumer Apps track (1st place) among 500+ participants

Experience

Finn (Airforge Tech Private Limited) **May 2026 – July 2026**
Software Engineering Intern *Bengaluru, India*

- Contributed **25+ merged pull requests** across **5+ production services**, including **AI voice agents**, backend APIs, authentication systems, WhatsApp automation services, deployment infrastructure, and customer-facing dashboards.
- Built **LLM-powered agent workflows**, **backend services, REST APIs**, and **event-driven workflows** powering real-time conversational AI and enterprise customer interactions in production.
- Managed **staging and production environments** with **twice-weekly deployments** using **Docker, Kubernetes, Nginx, and CI/CD pipelines**, improving release reliability and operational scalability.

UNSW Sydney **May 2026 – August 2026**
AI/ML Research Intern under Dr. Rohitash Chandra | GitHub *Sydney, Australia (Remote)*

- Contributed to a **published research paper** applying **SOMs** and **autoencoders** to hyperspectral satellite data for gold mineral exploration, fixing a **critical training bug** that raised **mineral recall from 0% to 76.6%**.
- Audited the full **unsupervised ML pipeline** and corrected **8 bugs** including incorrect **Cohen's** κ, mismatched autoencoder data masks, unseeded **PyTorch RNGs**, and rebuilt an irreproducible ground-truth raster from a **public government WFS API**.
- Designed an **evaluation framework** with **hyperparameter sweeps**, **McNemar significance tests**, **stability analysis** and **ablation studies**, proving **SOM outperforms k-means** on the target geological unit across all metrics.

Projects

PollForge | *Python, C#, Node.js, Kubernetes, Docker, Azure (AKS/ACR)* GitHub | Demo

- Built and deployed a **polyglot 5-service platform** on **Azure Kubernetes Service (AKS)**, provisioning **Azure Container Registry (ACR)** from scratch, with **StatefulSet**-backed **persistent storage** verified to survive live pod termination testing.
- Proved **zero-downtime, self-healing infrastructure** through deliberate **chaos testing**, and eliminated static cloud credentials via an **OIDC-authenticated CI/CD pipeline** shipping every commit to **ACR** and **AKS**, backed by **Prometheus/Grafana** observability and auto-renewing **HTTPS** via **cert-manager**.

DockHealer | *Python, Docker, Ollama (local LLM), Terraform, Azure, GitHub Actions, Trivy* GitHub

- Built a **local-LLM Dockerfile generator** whose output is verified deterministically: **16-rule validator** → real **docker build** → **hardened smoke-test container**; a **bounded repair loop** (22 failure classes, fingerprinting) **guarantees termination** and **stops for a human** on project-level faults.
- Provisioned an **Azure VM with Terraform** (NSG locked to one `/32`), **GitHub Actions CI**, **186 tests** via injected fakes, and a failure-injection **evaluation harness** (Wilson 95% CIs); logs **redacted and fenced as untrusted data** against prompt injection.

Helios | *Kubernetes, Go, ArgoCD, Argo Rollouts, Prometheus, Grafana, Kyverno, Docker, Azure, cert-manager* GitHub | Demo

- Built an **Internal Developer Platform** with a **self-healing GitOps pipeline** (GitHub Actions → ArgoCD) running **metrics-driven canary releases** that query live **Prometheus** signals mid-rollout and **auto-rollback** on degradation, with zero manual deployment steps.
- Wrote a custom **Go Operator (Kubebuilder)** reconciling a self-built **CRD** to provision isolated **per-PR preview environments** with finalizer-based cleanup and least-privilege RBAC; deployed on **k3s on Azure** with automated **TLS (cert-manager + Let's Encrypt)**, hardened via **Trivy, Kyverno, Falco**, and load-tested with **k6**.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C/C++, C#, Bash
Technologies: React.js, Next.js, Node.js, Express.js, Flask, Git, GitHub, MongoDB, PostgreSQL, Redis, WebSockets, REST APIs, GCP, Postman, Inngest, HTML/CSS, pytest
AI/ML & Data Science: LLMs, Ollama, NLP, LangChain, Hugging Face, FAISS, Scikit-learn, NumPy, Pandas
Web3: 0G, Uniswap, Privy
DevOps & Cloud: Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Helm, Azure, AKS, ACR, VNet/NSG, Linux, Nginx, Prometheus, Grafana, cert-manager, Trivy, OIDC Workload Identity Federation